

ECE 516: SYSTEM CONTROL ENGINEERING
Project Report

Design of an Automobile Cruise Control system

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Abstract—This project designs a state-feedback controller for a car cruise control system using control engineering principles. The open-loop transfer function was derived and analyzed to determine the desired closed-loop behavior. A desired characteristic equation was computed based on desired parameters, and a state-feedback controller was designed to achieve the targeted car dynamics.

1. Introduction

This project investigated the design of a cruise control system for a car, analyzing the combined effects of engine driving force and air drag. We developed a mathematical model of the car's dynamics that incorporated these essential forces. By applying control system engineering techniques, we designed a controller that analyzed the car's past and current speed and adjusted the engine torque to maintain a desired constant velocity. This project explored how modeling and control theory could be used to achieve precise cruise control functionality, even when accounting for the influence of air resistance.

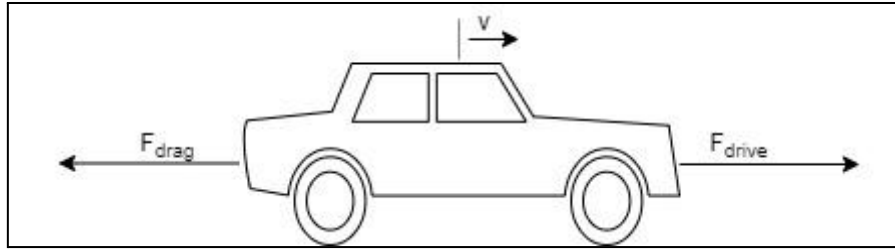


Figure 1. Forces acting on the vehicle.

2. Governing Equations

The equations of motion of the car we derived based on Newton's Second Law, "At any instant of time, the net force on a body is equal to the body's acceleration multiplied by its mass or, equivalently, the rate at which the body's momentum is changing with time". The friction at the driven wheel and rotational inertia were neglected for simplicity.

$$\Sigma F = m \cdot a(t) = F_{driving}(t) - F_{drag}(t); \quad (i)$$

$$= m \cdot \dot{v}(t) = F_{driving}(t) - C_D \cdot v^2(t); \quad (ii)$$

where, m = mass of the car

a = acceleration of the car

v = velocity of the car

C_D = coefficient of drag of the car

However, the force due to air resistance introduced a non-linearity in the system as it is proportional to the square of the car's velocity. To proceed, linearization was required.

2.1. Linearization

To linearize the system, Taylor series expansion was implemented to approximate the non-linear part of the function (v^2). Since this needs a point to expand the function around, an initial/pre-specified velocity (v_i , constant and not varying with time) was selected. In this initial state, without any disturbance, the driving force is exactly equal to the air resistance, and thus the system is in equilibrium (acceleration is zero). Substituting the same in equation (ii) we get,

$$0 = F_i - C_D \cdot v_i^2 \Rightarrow F_i = C_D \cdot v_i^2; \quad (iii)$$

With a change of variable, defining $v(t) = v_i + \delta v(t)$ and $F_{driving}(t) = F_i + \delta F(t)$ where δv and δF are time-varying deviations of the velocity and input (driving force) from the constant/preset equilibrium values.

$$v^2(t) \approx v_i^2 + 2 \cdot v_i \cdot \delta v(t) + h.o.t ;$$

$$\dot{v}(t) = \delta \dot{v}(t)$$

Substituting this linear approximation of v^2 in (iii),

$$\begin{aligned} m \cdot \delta \dot{v}(t) &= F_{driving}(t) - C_D \cdot \left[v_i^2 + 2 \cdot v_i \cdot \delta v(t) \right] ; \\ &= F_{driving}(t) - C_D \cdot v_i^2 - 2 \cdot C_D \cdot v_i \cdot \delta v(t) ; \\ &= \delta F(t) - 2 \cdot C_D \cdot v_i \cdot \delta v(t) ; \end{aligned} \quad (iv)$$

Where $\delta F(t) = F_{driving}(t) - F_i$, the deviation of the driving force from the value required to maintain constant speed at v_i . The system after linearizing becomes a relative system about the preset equilibrium point.

3. State Space Representation

Since this is a first-order system, the phase variable and observer canonical forms are the same. With vehicle velocity being the one state needed, we get the following matrices for the state space representation:

$$\begin{aligned} A &= \left[\frac{2 \cdot C_D \cdot v_i}{m} \right] & B &= \left[\frac{1}{m} \right] \\ C &= [1] & D &= 0 \end{aligned}$$

4. Open Loop Transfer function

4.1. Linearized equation in the frequency domain

To generate the state space representation and the transfer function, the equations in the time domain need to be converted into the frequency domain using the Laplace transform.

$$\mathcal{L}[\delta v(t)] = dV(s) ;$$

$$\mathcal{L}[\delta F(t)] = dF(s) ;$$

$$\therefore m \cdot s \cdot dV(s) = dF(s) - 2 \cdot C_D \cdot v_i \cdot dV(s) \quad (v)$$

4.2. Symbolic

From (v) the transfer function in its symbolic form is as follows:

$$\text{Transfer function, } G(s) = \frac{dV(s)}{dF(s)} = \frac{1}{m \cdot s + 2 \cdot v_i \cdot C_D}$$

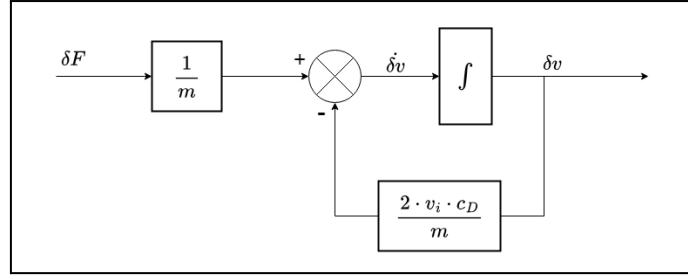


Figure 2. Simulation diagram showcasing the open-loop transfer function of the system.

4.3. With Parameters

For the mass and drag coefficient values, the ‘Pagani Huayra’ was chosen as a reference,

$$m = 1250 \text{ kg} \quad v_i = 90 \text{ mph} = 40.2336 \frac{\text{m}}{\text{s}} \quad c_d = 0.32$$

Substituting these values in the symbolic transfer function, we obtain:

$$\text{Transfer function, } G(s) = \frac{dV(s)}{dF(s)} = \frac{1}{1250 \cdot s + 24.9448}$$

This transfer function has a pole at $s = -\frac{24.9448}{1250}$.

5. Analysis of original system dynamics

5.1. Response to unit input

Comparing the transfer function obtained with the general transfer function of a first-order system, we can determine the values for the time constant (and hence the rise time) and amplitude.

$$G(s) = \frac{A}{\tau \cdot s + 1} = \frac{\frac{1}{24.9448}}{\frac{1250}{24.9448} \cdot s + 1}$$

From the above comparison,

$$\text{Time constant, } \tau = 50.11 \text{ s} \quad \text{Amplitude, } A = 0.0401 \frac{\text{m}}{\text{s}} \quad \text{Rise time} = 2.2 \cdot \tau = 110.242 \text{ s}$$

By taking inverse Laplace, the response of a system to a unit input will be of the form:

$$\delta v(t) = 0.0401 \cdot (1 - e^{-(t/50.11)})$$

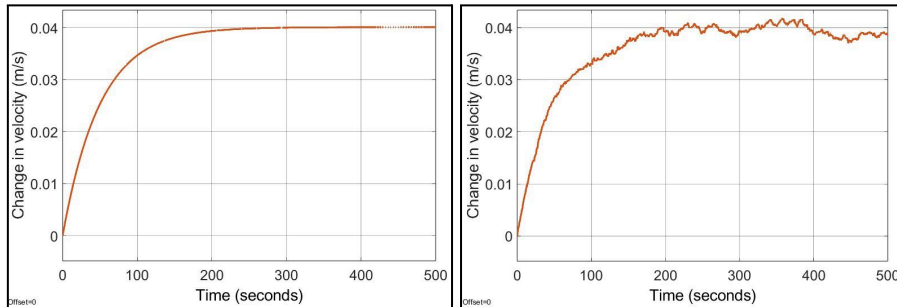


Figure 3. Response of the system to a unit input a) without noise and b) with noise.

Fig. 3a displays the response of the system to a unit step input with the steady state change in velocity around 0.04. Fig 3b displays the same response but with added noise simulating adverse road conditions.

5.2. Response to change in road gradient

Another form of external input that the cruise control system needs to counter is gradients in the road. When the gradient in the road changes, the horizontal component of the weight acts against (uphill) and in (downhill) the direction of the driving force.

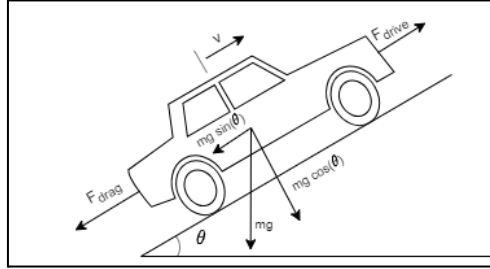


Figure 4. Free body diagram of a car on a positive gradient.

For small values of theta, the horizontal component, $m \cdot g \cdot \sin\theta$ can be linearized about zero where $\theta = 0 + \delta\theta$ and therefore $\sin\theta = \sin(0) + \cos(0) \cdot \delta\theta + h.o.t. = \delta\theta$. The horizontal component then becomes, $m \cdot g \cdot \delta\theta$. For a positive gradient, if the input force is not increased, it will result in deceleration of the vehicle and vice versa as shown in Fig. 5a and 5b.

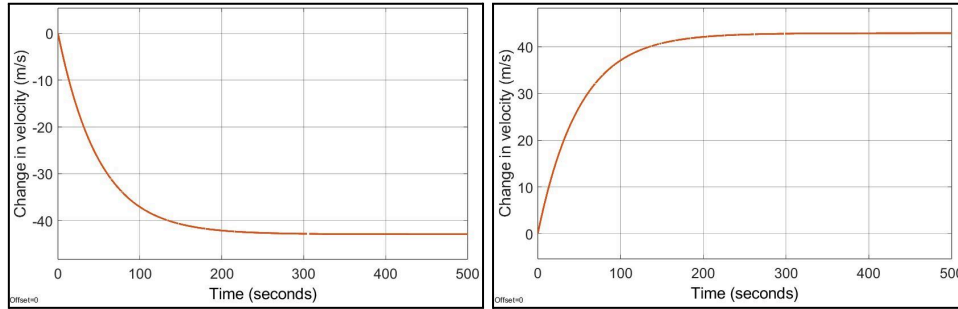


Figure 5. Response of the system to a change in gradient a) 5 deg uphill and b) 5 deg downhill.

5.3. Stability

The system has only one pole which lies at -0.0199. The absence of encirclements of the point (-1, 0), in the Nyquist plot indicates the stability of our system according to the Nyquist stability criterion. The root locus plot has 1 branch indicating the presence of only one pole. These plots also indicate the possible closed-loop poles, which can lie in the region enclosed by the ellipse and straight lines in the root locus plot, corresponding to a frequency greater than 0.18 for a desired rise time of less than 10 s, and a damping ratio of 0.06, for an overshoot less than 10%.

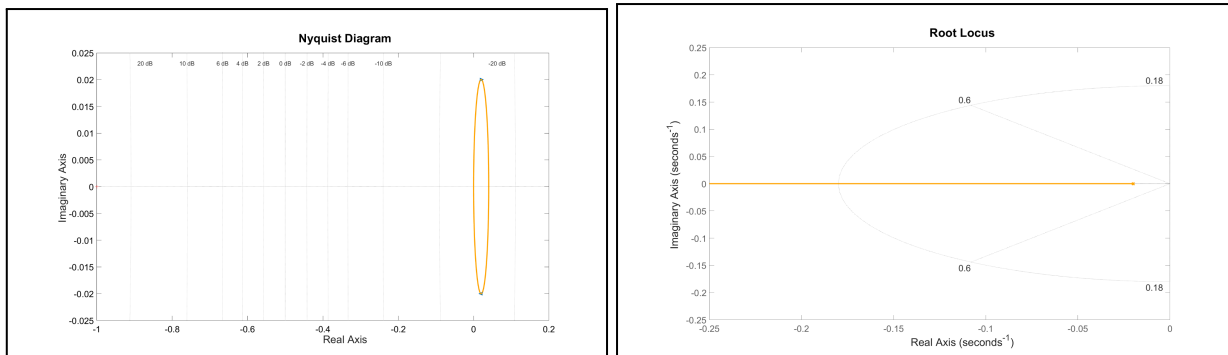


Figure 6. a) Nyquist and b) Root locus plots for the system

5.4. Controllability and Observability

Due to the system being first order, the controllability matrix equals $C_m = [B]$ and the observability matrix equals $\theta_m = [C]$. Since both the matrices are already non-singular, the system is both controllable and observable.

6. Designing Controllers

6.1. State Feedback Controller - Desired system dynamics

By introducing state feedback and changing the value of the gains, the placement of the poles of the system can be controlled. This gain can be engineered to augment the behavior of the system.

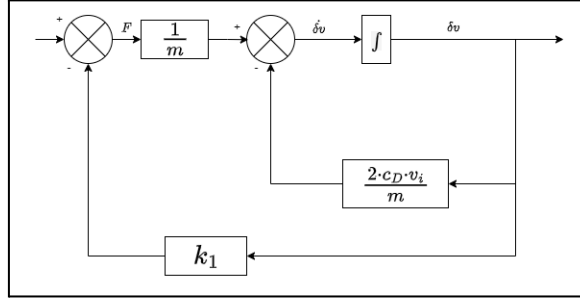


Figure 7. Simulation diagram for the system with state feedback.

If state-feedback with a gain of k_1 is introduced (Fig. 7), the transfer function of the system changes to

$$G(s) = \frac{1}{m \cdot s + (2 \cdot v_i \cdot c_d + k_1)} = \frac{1}{(2 \cdot v_i \cdot c_d + k_1)} \cdot \frac{1}{\frac{m}{(2 \cdot v_i \cdot c_d + k_1)} \cdot s + 1}$$

Comparing this to the general transfer function for a first-order system the new values for the time constant and amplitude for unit input are:

$$\tau = \frac{m}{(2 \cdot v_i \cdot c_d + k_1)} \quad A = \frac{m}{(2 \cdot v_i \cdot c_d + k_1)}$$

The time constant (τ) and rise-time ($2.2 \cdot \tau$) represent the speed of the response of a system to an input or disturbance. If it is too high, the system will slow to respond to disturbances (which is desirable) but also to inputs (undesirable). Therefore it is crucial to choose the value correctly since it decides how quickly the system reacts to an uneven road or change in road gradient. If it's too small, it can affect the comfort of the passengers. As a result, a desired time constant of 10s was chosen, putting the time constant at 4.54s and the state-feedback gain at 250.055. The desired transfer function then is:

$$G^*(s) = \frac{1}{1250 \cdot s + 275}$$

Putting the new pole at $s = -0.220$. From the results in Fig 8. for unit-step (a), uphill (b), and downhill (c) performance, it is visible that the changes in velocity are far reduced with the state feedback provided. The response is quicker compared to the open-loop response as well.

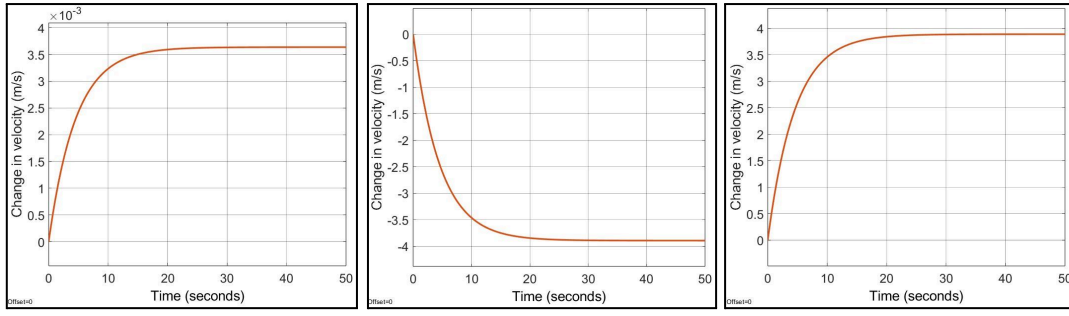


Figure 8. Response of the system to a) a unit step, b) 5 deg uphill gradient, and c) 5 deg downhill gradient with state feedback.

6.2. PID Controller

A PID control is used in the system to get a stable system with minimum steady-state gain.

The overall control function of the systems is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

where, K_p = Proportional Gain,

K_i = Integral Gain,

K_d = Derivative Gain

As a starting point for PID gains, we kept the proportional gain, K_p , the same as state feedback gain, i.e $K_p = 250.055$. This was the reference point for us to calculate the K_i and K_d values. By using trial and error methods, we were able to pinpoint the gains. We manually adjusted the gains further to obtain a stable graph.

1. Controller Transfer Function:

$$\text{Transfer function, } G_c(s) = K_p + \frac{K_i}{s} + K_d s$$

2. Plant Transfer Function (Closed Loop):

$$\text{Transfer function, } G(s) = \frac{dV(s)}{dF(s)} = \frac{1}{m \cdot s + 2 \cdot v_i \cdot c_d}$$

3. Transfer Function of the whole system is given by:

$$\begin{aligned} G_c(s) * G(s) &= (K_p + \frac{K_i}{s} + K_d s) \cdot \frac{1}{m \cdot s + 2 \cdot v_i \cdot c_d} \\ &= \frac{s^2 K_d + s K_p + K_i}{m s^2 + 2 \cdot v_i \cdot c_d s} \end{aligned}$$

Conventional PID controllers utilize proportional and integral control to address the current error and its historical accumulation, respectively. However, for a cruise control system in a dynamic environment, these factors might not be sufficient. The derivative term, as expressed mathematically, focuses on the rate of change

of the error signal. This allows the controller to anticipate future changes in the car's speed caused by factors like road grade or varying air resistance. Even though we might maintain a constant reference speed (desired error) during the experiment, real-world driving scenarios involve continuous fluctuations in the actual speed. The derivative control term plays a crucial role in counteracting these transient disturbances and ensuring the car's speed remains close to the set point.

We have defined the gains for the system as follows:

Table 1. Gains selected for the PID controller.

Gain	Final Value
K_p	250.0522
K_i	101.95
K_d	50
Filter coefficient (N)	100

Below is the Simulink Model of the system with the system with both PID controller and state feedback with a switch to change between the two:

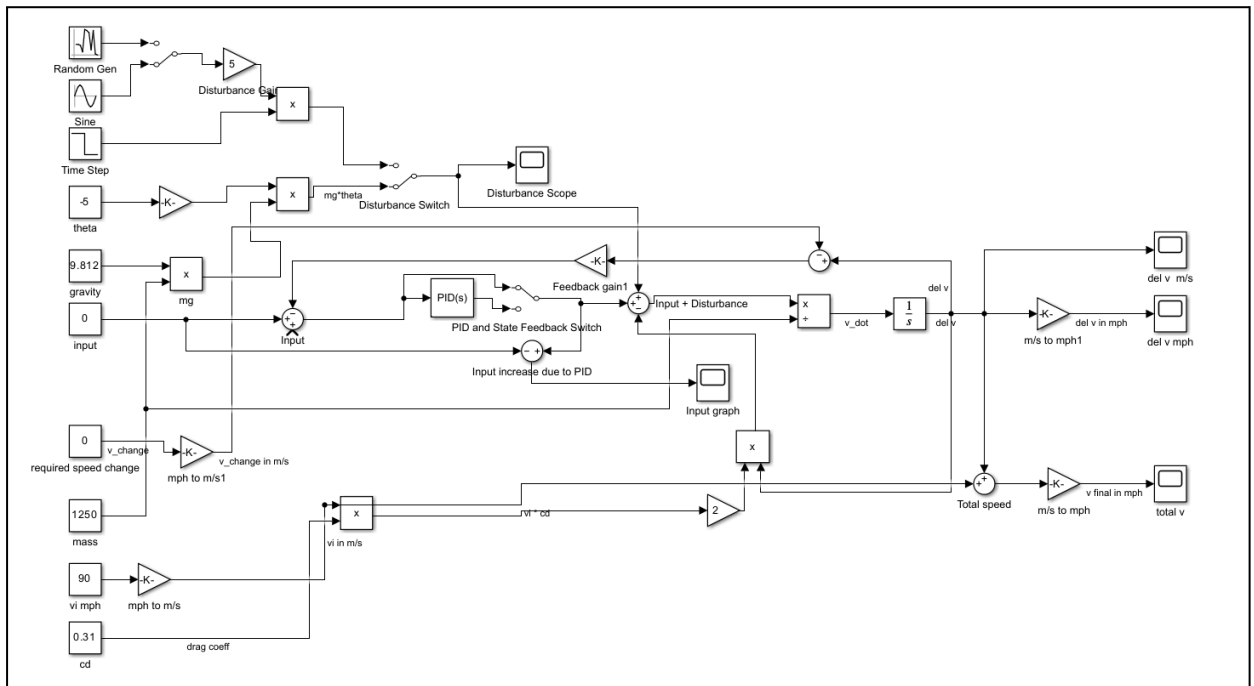


Figure 9. Simulink model of the system with disturbances and both controllers.

Fig. 10 displays the response of the system with the PID controller to a unit input (a) and uphill (b) and downhill (c) gradients of 5 degrees.

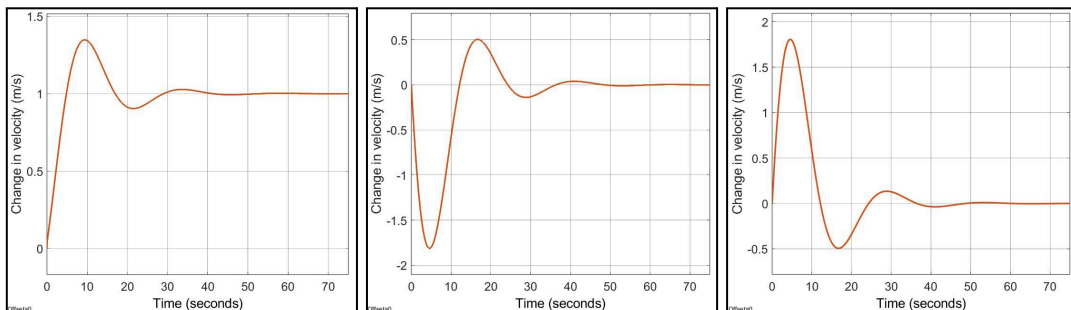


Figure 8. Response of the system to a) a unit step, b) 5 deg uphill gradient, and c) 5 deg downhill gradient with the PID controller.

7. Conclusion:

This project delved into the design of a cruise control system for a car, utilizing the control theory principles explored in our coursework. We encountered several challenges throughout the process, including obtaining a suitable transfer function, linearizing the nonlinear equations due to air drag, and finally designing a PID controller. Overcoming these challenges provided valuable insights into the practical application of control systems. While initial overshoot issues were encountered during the early stages of controller design, we are pleased to report the successful development of a PID controller for the cruise control system.

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